Supplementary Step 9 Visualization Report

Overview

Metric	Value
Included participant-sessions	36
Included questions	30
Included trials	1075
Primary block-adjusted condition p-value	0.184407
Condition x block-order interaction p-value	0.994710
Condition x within-block-position interaction p-value	0.205681
Condition x condition-order interaction p-value	0.594901
Early-vs-late interaction p-value	0.916963
Back ROI benchmark p-value	0.429482
Fixed-window benchmark p-value	0.276214

Interpretation note. In the block-adjusted mixed-effects analysis, the abstract-versus-concrete effect in the left frontal ROI was reduced and was not statistically significant after controlling for global block order, within-block trial position, participant condition order, question length, and item difficulty (beta=3.020e-08, SE=2.276e-08, p=0.184407, 95% CI [-1.440e-08, 7.481e-08]). This suggests that the residual frontal tendency does not survive clear statistical support once block structure is modeled explicitly. None of the predeclared condition-by-block interaction terms reached the alpha=0.05 threshold.

Descriptive note. These figures focus on how the realized block structure, within-block position, and participant condition order relate to the front ROI outcome before and after block adjustment. Mean front-ROI values by session half were: Early blocks (1-5) / Abstract: -3.784e-08; Early blocks (1-5) / Concrete: -5.817e-08; Late blocks (6-10) / Abstract: -2.493e-08; Late blocks (6-10) / Concrete: -3.814e-08.

Core Step 9 Diagnostic Figures

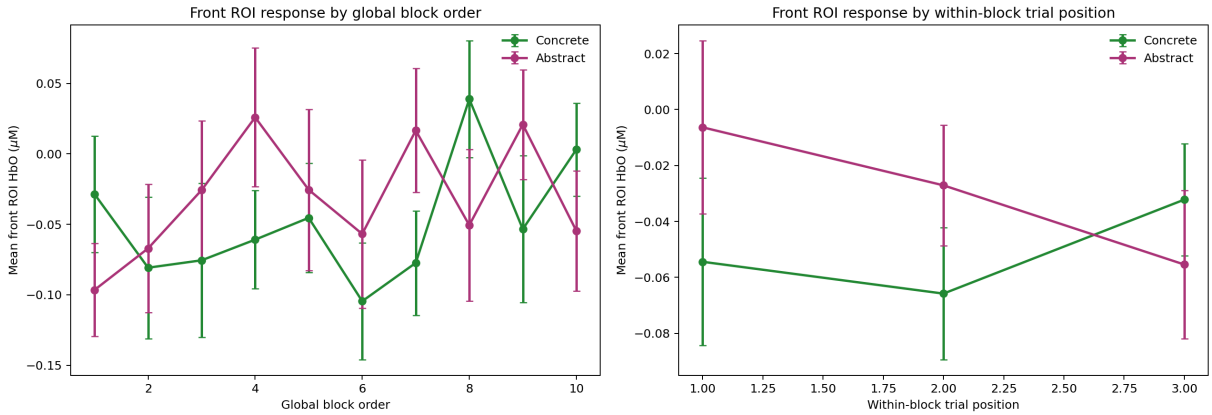

Figure 1: Front ROI response as a function of realized block order and within-block question position.

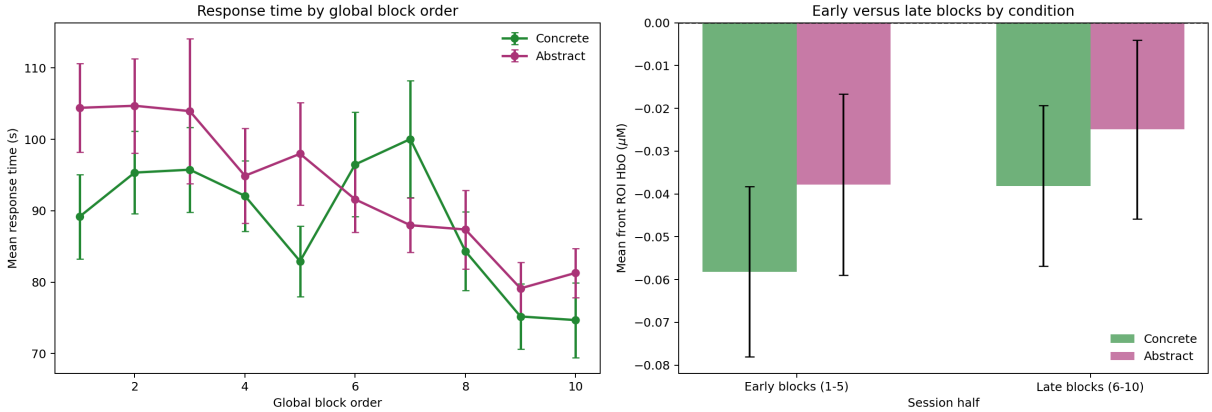

Figure 2: Response-time progression across the session and the early-versus-late condition comparison.

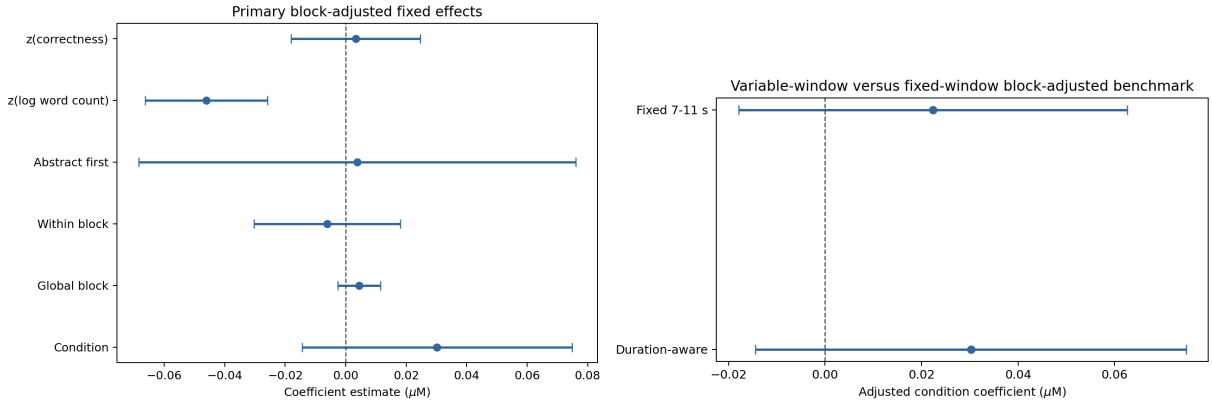

Figure 3: Primary block-adjusted coefficient summary and the duration-aware versus fixed-window benchmark.

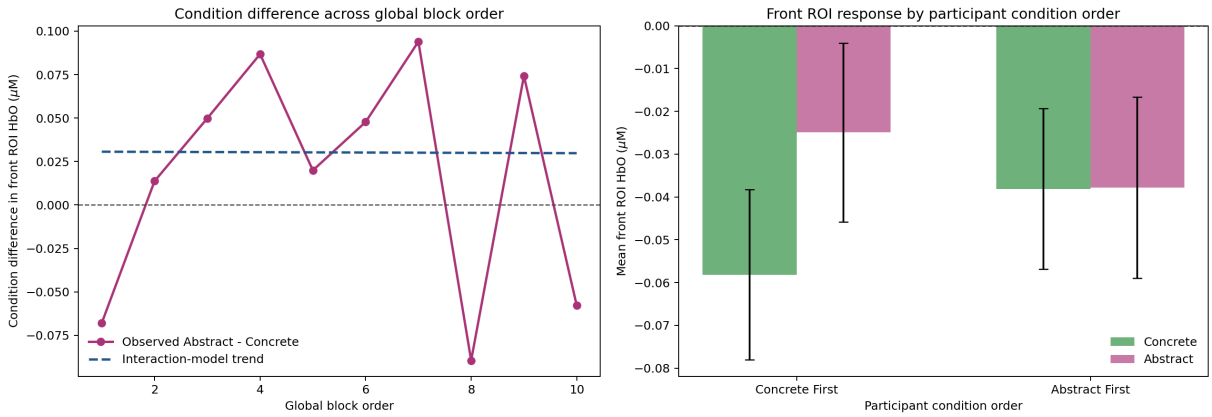

Figure 4: Secondary interaction-focused plots for global block order and participant condition order.

Additional Step 9 Visualization Diagnostics

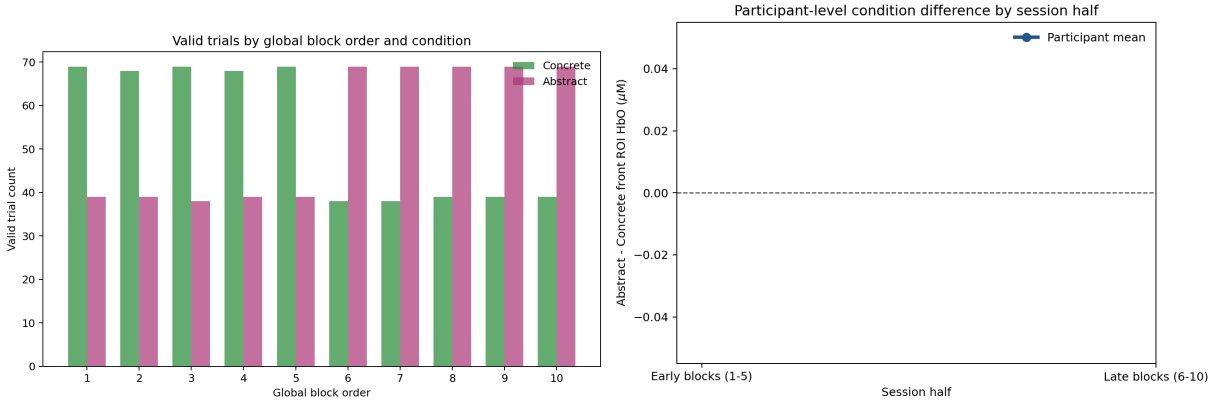

Figure 5: Additional descriptive views: valid trial counts by block and condition, and participant-level abstract-minus-concrete front-ROI differences from early to late session halves.